

Elements of Computer Science: Database Assignment 6

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Vinay Aditya Thirunagari

Ayush Ramkaran Yadav

1.

a. $\pi_{\{Price\}}(\sigma_{\{Name = 'table top' \wedge Weight = 20\}}(Intermediate_Product))$

b. $\pi_{\{PNR, Name\}}($
 $Final_Product \triangleright \triangleleft_{\{Final_Product.PNR = stored_in.PNR\}}$
 $\sigma_{\{Quantity > 100\}}(stored_in)$
 $)$

c. $\pi_{\{Name\}}($
 $(Intermediate_Product \triangleright \triangleleft_{\{Intermediate_Product.PNR =$
 $consists_of.PNR2\}}consists_of)$
 $\cup$
 $(Final_Product \triangleright \triangleleft_{\{Final_Product.PNR = consists_of.PNR2\}}$
 $consists_of)$
 $)$

d. $\pi_{\{PNR, Name\}}($
 $Final_Product \triangleright \triangleleft$
 $\pi_{\{S1.PNR\}}($
 $\sigma_{\{S1.PNR = S2.PNR \wedge S1.WID \neq S2.WID\}}($
 $\rho_{\{S1\}}(stored_in) \times \rho_{\{S2\}}(stored_in)$
 $)$
 $)$
 $)$

e. $AllW := \pi_{\{WID, Address, Capacity\}}(Warehouse)$

Smaller :=

$\pi_{\{W1.WID, W1.Address, W1.Capacity\}}($
 $\sigma_{\{W1.Capacity < W2.Capacity\}}$
 $(\rho_{\{W1\}}(Warehouse) \times \rho_{\{W2\}}(Warehouse))$
 $)$

Result :=
AllW \ Smaller

$\pi_{\{WID, Address\}}(Result)$

- f. AllProducts := $\pi_{\{PNR\}}(Intermediate_Product) \cup \pi_{\{PNR\}}(Final_Product)$
 AllProdRenamed := $\rho_{\{PNR1\}}(AllProducts)$
 R := $\pi_{\{PNR2, PNR1\}}(consists_of)$
 IntermediatesInAll := $R \div AllProdRenamed$
 $\pi_{\{PNR, Name\}}(Intermediate_Product \triangleright \triangleleft_{\{PNR=PNR2\}}$
 IntermediatesInAll)
 g. CI := $consists_of \triangleright \triangleleft_{\{consists_of.PNR2 = Intermediate_Product.PNR\}}$
 Intermediate_Product
 TwoIntermediates := $\rho_{\{PNR \leftarrow PNR1\}}(\pi_{\{C1.PNR1\}}(\sigma_{\{C1.PNR1 = C2.PNR1 \wedge C1.PNR2 \neq C2.PNR2\}}(\rho_{\{C1\}}(CI) \times \rho_{\{C2\}}(CI)))$
)
)

StockedFinals := $\pi_{\{PNR\}}(Final_Product \triangleright \triangleleft_{\{Final_Product.PNR = stored_in.PNR\}}(stored_in))$
 Result := StockedFinals \ TwoIntermediates

$\pi_{\{PNR\}}(Result)$

2.

- a. AnnaCustomers := $\sigma_{\{FirstName = 'Anna'\}}(Customer)$

AnnaOrders := $Order \triangleright \triangleleft_{\{Order.CNR = AnnaCustomers.CNR\}}$
 AnnaCustomers

PerAnna := $\gamma_{\{CNR, totalQty \leftarrow SUM(Quantity)\}}(AnnaOrders)$

Result := $\gamma_{\{avgQty \leftarrow AVG(totalQty)\}}(PerAnna)$

$\pi_{\{avgQty\}}(Result)$

- b. $\pi_{\{totalStock\}}(\gamma_{\{totalStock \leftarrow SUM(Stock)\}}(\sigma_{\{Price < 50\}}(Product)))$

- c. CheapestPrice :=

$\gamma_{\{\min \leftarrow \text{MIN}(\text{Price})\}}(\text{Product})$

CheapestProd :=

$\sigma_{\{\text{Price} = \min\}}(\text{Product} \times \text{CheapestPrice})$

CPOrders :=

$\text{Order} \triangleright \triangleleft_{\{\text{Order.PNR} = \text{CheapestProd.PNR}\}} \text{CheapestProd}$

MaxQ :=

$\gamma_{\{\max Q \leftarrow \text{MAX}(\text{Quantity})\}}(\text{CPOrders})$

Result :=

$\pi_{\{\text{CNR}\}}(\sigma_{\{\text{Quantity} = \max Q\}}(\text{CPOrders} \times \text{MaxQ}))$

d. AvgP :=

$\gamma_{\{\text{avgP} \leftarrow \text{AVG}(\text{Price})\}}(\text{Product})$

$\pi_{\{\text{Name}\}}(\$

$\sigma_{\{\text{Product.Price} > \text{AvgP.avgP}\}}(\$

$\text{Product} \times \text{AvgP}$

$)$

$)$